ICTCE04

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Weekly Progress Report Form

Name - Mr. Sai Swam Wan Hline, Student ID - 6609216

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Company Name /Employer Name - **HTZKKL Technologies Co.,Ltd.**

Reporting Period: *Week No.2, September 2026, 9/9/2026 - 15/9/2026, Total 40 Hours*

1. Name of project/task/issue/topic of student's responsibility.

(List each of the performed tasks.)

Project: Multi-Tenant School Intranet & Academic Management Platform

Task 1: Design and implementation of the Examinations & Grading Module (exam timetables, mark entry, grade boundaries, and student report card generator).

Task 2: Development of the School Finance & Payroll System (fee structures, multi-line invoices, payment slip verification, and teacher payslips).

Task 3: Implementation of the Academic Promotions Engine for class-to-class student progression and academic year rollovers.

Task 4: Integration of automated messaging bots (Viber Bot & Telegram Bot) for instant parent alerts (exam results, attendance, fee reminders).

Task 5: Security hardening (race-safe document numbering, IDOR remediation, magic-byte file validation) and construction of an automated 56-test regression suite.

2. Summary of the performed tasks.

(Describe the type of work that the student has accomplished.)

During the second week of the project:

Examinations & Report Cards: Built the end-to-end examination pipeline (app/modules/examinations). Implemented exam timetable scheduling, class-based

mark entry with validation limits, dynamic GPA/grading calculation, and printable A4 student report card generation. Built corresponding frontend pages at /dashboard/examinations and /dashboard/results.

Finance, Invoicing & Payroll: Developed the financial operations module (app/modules/finance). Structured multi-item fee invoices, tuition fee schedules, manual payment reconciliation with bank slip upload verification, and staff payroll generation. Designed printable official payment receipts and staff payslip components.

Academic Student Promotions: Implemented the batch promotions system (app/modules/promotions) enabling school administrators to promote, retain, or transfer entire cohorts between grade levels at the end of an academic term, with full audit trail logging and rollback capability.

Viber & Telegram Bot Integrations: Created dedicated webhook handlers and API clients for Telegram (app/modules/telegram) and Viber (app/modules/viber). Automated parent notifications for published exam grades, unexcused student absences, and outstanding invoice reminders with secure pairing codes.

Security Hardening & Automated Testing:

Replaced naive database autoincrement numbering with concurrency-safe document counters for invoices, receipts, and payslips to prevent race conditions.

Enforced file upload security by validating MIME types using magic-byte headers rather than client-supplied extensions.

Built a comprehensive 56-test automated regression suite with pytest, verifying multi-tenant data boundaries, authorization checks, and calculation integrity with a 100% pass rate.

3. Application of concepts, theories, knowledge, or experiences that applied to cooperative education.

Financial Data Integrity & Idempotency: Applied ACID database transaction principles to billing and payments. Designed payment verification to be strictly idempotent, ensuring simultaneous requests cannot cause double-crediting or duplicate receipt issuance.

Concurrency & Distributed Race Conditions: Applied atomic database row-locking (SELECT FOR UPDATE / sequence isolation) to guarantee sequential document numbering under concurrent operations.

Defensive Web Security: Applied OWASP Top 10 guidelines to eliminate Insecure

Direct Object References (IDOR) on financial slips and implemented server-side magic-byte file inspection to prevent malicious file execution.

Event-Driven Webhook Architectures: Applied asynchronous HTTP client patterns and webhook security protocols (verifying HMAC signatures and webhook secrets) for external third-party communication APIs (Telegram & Viber).

Automated Test-Driven Verification (TDD): Applied software quality assurance methodologies by writing comprehensive unit and integration tests using Pytest and FastAPI TestClient.

4. Problems/obstacles encountered during the work.

(Provide details on how to solve the problems/obstacles.)

Problem 1: Concurrent Document Number Generation Collisions:

Detail: Under concurrent invoice and receipt generation, multiple transactions read the same latest document number, leading to duplicate invoice numbering collisions (e.g. two separate parents receiving INV-2026-0001).

Solution: Implemented dedicated transactional counter tables with atomic updates and database locks within a nested savepoint, guaranteeing strictly unique, contiguous document numbering.

Problem 2: Cross-Tenant Information Leakage via Document Lookups (IDOR):

Detail: During testing, an administrator from School A could potentially access payment receipts or exam results from School B if they knew the document ID.

Solution: Added strict tenant-scoping checks inside verify_invoice_access and verify_exam_access dependencies, validating that document.school_id == current_user.school_id and returning 404 Not Found if the tenant IDs do not match.



(……Nay Min.....)

Job Supervisor

Date ……15/9/2026.....



(….Sai Swam Wan Hline.....)

Student

Date ……15/9/2026.....